

AI201: Introduction to Machine Learning

Course Project Guidelines

Monsoon 2026

1 Purpose and Structure

The course project is intended to provide hands-on experience in applying machine learning to a meaningful problem. The emphasis is on **problem formulation, methodological correctness, experimental design, evaluation, analysis, and communication**, rather than on using the most complex model.

A strong project should demonstrate that the team can:

- formulate a meaningful problem as a machine-learning task;
- understand and analyse an appropriate dataset;
- establish suitable baselines;
- select and justify ML methods;
- use correct training, validation, and evaluation procedures;
- compare alternatives experimentally;
- analyse errors, limitations, robustness, or subgroup behaviour; and
- communicate and reproduce the work clearly.

Model complexity itself will not be rewarded. A carefully designed project using relatively simple methods can receive full marks if the methodology, experiments, and analysis are strong.

Project weight: 20% of the course grade

Group size: 3–4 students

Component	Marks
Project Proposal	5
Mid-semester Progress Report	2.5
Mid-semester Presentation	5
End-semester Final Report	2.5
End-semester Presentation	5
Total	20

The project is carried out in groups, but individual marks may differ. Intra-group evaluation, demonstrated contribution, and individual understanding during presentations or discussions may be used to determine individual marks.

Presentation instructions and format will be provided separately.

2 Choosing a Project

Application-oriented projects are strongly encouraged. Pick a meaningful application and investigate how machine-learning methods can be applied effectively and responsibly.

Suitable application areas include, but are not limited to:

- healthcare and public health;
- education;
- sustainability, environment, and climate;
- agriculture;
- accessibility;
- transportation and urban systems;
- finance and commerce;
- computational social science;
- text, image, audio, or scientific data analysis;
- forecasting, recommendation, clustering, and anomaly detection.

Projects related to **AI for Social Good** are particularly welcome where there is a meaningful societal problem and appropriate data. However, the social importance of an application does not substitute for rigorous ML methodology.

Other project types are possible with instructor approval:

- **Algorithmic project:** develop or meaningfully modify an ML method and evaluate it.
- **Reproduction/extension project:** reproduce previous work and then add substantial new experiments, analysis, baselines, datasets, ablations, or extensions.
- **Theoretical project:** analyse or prove non-trivial properties of an ML method. Purely theoretical projects require prior approval and are not the primary focus of this course.

Publication-quality work is welcome, but **novel publishable research is not required**. Course grading is based on the quality of the ML work, not on whether the project produces a new state-of-the-art result.

Scope

A suitable project should have:

- a clear ML task or research question;
- sufficient but manageable data;
- a meaningful evaluation procedure;
- at least one appropriate baseline;
- enough scope for multiple experiments or comparisons;
- opportunities for meaningful analysis beyond reporting a single score; and
- feasible computation and workload for a team of 3–4 students.

Projects that consist primarily of running one standard classifier, reproducing a public notebook, prompt engineering, calling an external LLM/API, or training an unnecessarily large model will generally not provide sufficient scope.

Similarly, avoid projects whose main effort is collecting a new dataset or building substantial software infrastructure rather than doing machine learning.

3 Datasets and Data Understanding

Existing public datasets are strongly preferred. Dataset collection and curation can consume a large fraction of the semester without contributing directly to the ML learning objectives.

Useful starting points include:

- **Kaggle Datasets:** <https://www.kaggle.com/datasets>
- **UCI Machine Learning Repository:** <https://archive.ics.uci.edu/>
- **OpenML:** <https://www.openml.org/>
- **Hugging Face Datasets:** <https://huggingface.co/datasets>
- **Open Government Data Platform India:** <https://data.gov.in/>
- **Google Dataset Search:** <https://datasetsearch.research.google.com/>
- Dataset repositories associated with published papers, research laboratories, government agencies, or established benchmarks.

UCI and OpenML are particularly useful for classical ML problems, while Kaggle provides a broad collection of application datasets. Hugging Face can be useful for text, image, audio, and multimodal datasets. Government and research repositories may be especially appropriate for socially relevant applications.

Students are responsible for checking the **source, licence, intended use, documentation, and any privacy or ethical restrictions** associated with a dataset.

Even when a prepared dataset is used, students must conduct their own data understanding and exploratory analysis. Depending on the problem, this may include:

- size, attributes, and target variables;
- feature and target distributions;
- missing values and duplicates;
- class imbalance;
- correlations or dependencies;
- possible data leakage;
- subgroup representation; and
- potential biases or limitations of the data.

Collection of a new dataset should normally be undertaken only when it is necessary for the proposed problem and should be discussed with the teaching team early.

4 Expected Machine-Learning Methodology

A typical project should progress through

Problem $\rightarrow$ *Data* $\rightarrow$ *Baseline* $\rightarrow$ *Models* $\rightarrow$ *Evaluation* $\rightarrow$ *Analysis* $\rightarrow$ *Conclusions*.

Students should justify important choices including:

- preprocessing and feature representation;
- baseline selection;
- model selection;
- train/validation/test splitting;
- hyperparameter-selection procedure;
- evaluation metrics; and
- experimental comparisons.

Every project should normally include at least one meaningful baseline. Depending on the task, this may be a majority/random predictor, simple heuristic, linear or logistic regression model, decision tree, or an established domain baseline.

Correct experimental practice is essential. In particular:

- do not use the test set for model or hyperparameter selection;
- prevent data leakage;
- use suitable metrics rather than accuracy by default;
- compare models under consistent experimental conditions; and
- make experiments reproducible.

Where appropriate, projects should include analyses such as:

- ablation studies;
- sensitivity analysis;
- subgroup analysis;
- robustness analysis;
- qualitative or quantitative error analysis.

A final result such as “Model A obtains 92% accuracy” is not sufficient by itself. Students should investigate **why methods work, where they fail, and what can reasonably be concluded from the experiments**.

Methods taught in AI201 should play a meaningful role in the project. Methods beyond the syllabus are allowed when justified, but sophisticated external models should not replace understanding of the ML methodology.

Deep Learning, Pretrained Models, and Generative AI

Deep learning is permitted where appropriate but is neither required nor automatically rewarded.

If pretrained models, embeddings, or representations are used, students should document:

- the source/model used;

- what was pretrained and, where known, on what data;
- which components are frozen, fine-tuned, or trained;
- why the pretrained representation is appropriate;
- comparison with a meaningful baseline; and
- important limitations.

Generative AI tools may be used subject to the course AI-use and academic-integrity policies. However, a project whose principal technical methodology consists only of prompting or calling a generative-AI API will normally not constitute sufficient machine-learning work for AI201.

Students remain responsible for all generated code, references, claims, explanations, and experimental results.

5 Project Deliverables

All written project submissions must use the **AAAI paper format** available at [AAAI 2026 Press Formatting Instructions for Authors using L^AT_EX](#). The same document should evolve across the semester rather than being rewritten independently at every stage.

Proposal — 5 marks

The proposal should establish that the project is meaningful and feasible. It should contain:

- project title and team members;
- problem statement and motivation;
- precise ML task/research question;
- dataset and source;
- initial data understanding;
- proposed methods and baseline(s);
- planned evaluation metrics and validation strategy;
- brief related work;
- expected challenges and work plan; and
- proposed division of responsibilities.

Mid-semester Progress Report — 2.5 marks

The report should update the proposal and document actual progress. It should include:

- any revision to the original problem formulation;
- completed data analysis and preprocessing;
- implemented baseline(s);
- methods attempted;
- preliminary experimental results;

- difficulties, failed approaches, or open questions; and
- concrete plan for the remainder of the semester.

Final Report — 2.5 marks

The final report should present the complete project as a coherent technical paper. A typical structure is:

Abstract; Introduction; Related Work; Dataset; Methodology; Experimental Setup; Results; Error/Failure Analysis; Limitations and Responsible Use; Conclusion; Individual Contributions; References.

The final submission should make a clear distinction between observations supported by experiments and claims that remain speculative.

Code and Reproducibility

The final project submission must include code with sufficient documentation to reproduce the principal experiments. It should normally contain:

- a README;
- dependency/environment information;
- instructions for obtaining the data;
- preprocessing, training, and evaluation code;
- relevant configurations/hyperparameters; and
- instructions for reproducing important reported results.

A collection of undocumented notebooks with unclear execution order is not considered adequate documentation.

6 Evaluation, Team Contribution, and Academic Integrity

Projects will be evaluated primarily on:

1. **Technical and methodological quality:** Is the ML formulation sound? Are the methods appropriate? Are training, validation, evaluation, and comparisons correct?
2. **Scope and contribution:** Is there sufficient substantive work for the team size? Does the project go beyond reproducing an existing paper, notebook, or implementation?
3. **Analysis and insight:** Does the team understand the observed results? Are error analysis, comparisons, limitations, failures, and other observations meaningful?
4. **Communication and reproducibility:** Are the problem, methods, experiments, and conclusions presented clearly? Can the project be reproduced and defended by the team?

Negative results are acceptable. A carefully designed experiment showing that an approach does not work can be valuable when the failure is analysed and communicated appropriately.

All group members are responsible for understanding the complete project. Work may be divided, but each student should be able to explain the problem, data, methodology, evaluation, major results, and limitations.

A short contribution statement must be included in the final report. Confidential intra-group evaluation may also be used. Consequently, students in the same group need not receive identical marks.

The IIIT-Delhi Policy for Academic Misconduct applies fully to project work. Students must properly acknowledge borrowed ideas, text, code, figures, datasets, and other material. Existing work may be used only with appropriate attribution and in accordance with the course and institute policies.

Use of generative AI or coding assistants does not transfer responsibility for submitted work. Students must be able to explain and defend anything included in their submission.

7 From Course Project to Research

The primary objective of the project is learning; publication is **not** a grading requirement.

Nevertheless, particularly strong projects may provide a starting point for continued research after the course. Application-oriented projects may be relevant to venues and tracks concerned with real-world AI applications, including areas such as AI for Social Good, AI and Health, Human-Centered AI, and related workshops or conferences.

For example, the IJCAI-ECAI 2026 **AI and Social Good** track considered work in areas including healthcare, education, environmental sustainability, public health, urban planning, fairness, poverty alleviation, human rights, and vulnerable communities. Such submissions were expected to demonstrate both strong scientific quality and meaningful engagement with the real-world problem.

Students interested in developing their project toward an IJCAI submission should note that conference deadlines typically occur well before the conference itself. For reference, IJCAI-ECAI 2026 used:

- Abstract deadline: **12 January 2026**
- Full-paper deadline: **19 January 2026**

The official IJCAI 2027 submission dates have not yet been announced at the time these guidelines were prepared. Interested teams should therefore monitor:

<https://www.ijcai.org/>

and, as a practical research target, aim to have a substantially complete manuscript and experiments by **early January 2027** if they intend to pursue the next IJCAI cycle.

Students considering publication should discuss this with the instructors early. A publishable project usually requires substantially more work than is required for obtaining full marks in AI201, including stronger novelty, broader experiments, deeper analysis, and—for application-focused work—appropriate domain expertise or stakeholder engagement.

In short, a strong AI201 project has:

**a clear problem + appropriate data + a meaningful baseline + justified methods
+ correct evaluation + insightful analysis + reproducible experiments.**

8 Sample Project Directions

The following are illustrative project directions intended to help you think about suitable scope and problem formulations. These are **not ready-made project specifications**; each team must define its own problem, dataset, baselines, methodology, evaluation strategy, and analysis.

- **Student Performance Prediction:** Predict course outcomes or dropout risk using demographic, behavioural, or academic features; compare simple baselines with linear, tree-based, or ensemble models and include subgroup or error analysis.
- **Air Quality or Pollution Forecasting:** Predict pollutant levels using weather, temporal, and environmental variables; compare regression models and analyse where forecasting performance degrades.
- **Medical Risk Prediction:** Use a public clinical dataset to predict disease risk, readmission, or treatment outcome; emphasize class imbalance, calibration, suitable metrics, and responsible interpretation.
- **Text Classification:** Study problems such as sentiment, toxicity, spam, or misinformation detection; compare classical text representations with learned embeddings or pretrained representations and perform detailed error analysis.
- **Credit Risk or Default Prediction:** Predict financial risk using tabular data; compare interpretable and ensemble models and analyse important features, errors, and subgroup performance where appropriate.
- **Image Classification with Classical vs. Learned Features:** Compare hand-crafted or dimensionality-reduced features with pretrained image representations on a manageable public dataset.
- **Energy Demand or Consumption Forecasting:** Forecast electricity demand or building energy usage; compare simple baselines with ML regressors and analyse temporal, seasonal, or extreme-event failure cases.
- **Recommendation Systems:** Compare popularity-based, content-based, and collaborative-filtering approaches for movies, books, products, or similar data; evaluate using suitable recommendation metrics rather than a single aggregate score.
- **Anomaly Detection in Sensor or Network Data:** Detect unusual events using statistical, clustering, or representation-based methods; investigate the trade-off between false alarms and missed anomalies.
- **AI for Social Good Application:** Use an existing public dataset related to education, agriculture, accessibility, public health, sustainability, or urban systems; formulate a clear ML task and include analysis of limitations and potential societal impact.

9 Frequently Asked Questions

Q1. Can we use deep-learning models?

Yes. Deep-learning models are allowed when they are appropriate for the problem. However, using deep learning is **not required and will not receive additional credit by itself**. Projects should still include meaningful baselines, sound validation, appropriate evaluation, and analysis of results. A simpler model studied carefully can receive higher marks than a complex neural model used without adequate justification or analysis.

Q2. Will GPU resources be provided by the course?

No dedicated GPU resources will be provided by the course. Students should primarily plan experiments around resources available through platforms such as **Google Colab** and any GPU access normally available with such services.

The availability or use of a GPU will **not have any bearing on project assessment**. Projects should therefore be scoped so that the principal experiments can be completed with reasonable computational resources. Lack of access to high-end GPUs is not a disadvantage for grading.

Q3. Can we use pretrained models?

Yes. Pretrained models, embeddings, and feature representations may be used where appropriate. However, students must clearly state:

- the model or representation used and its source;
- what is pretrained and what is trained or fine-tuned by the team;
- why its use is appropriate for the problem;
- what meaningful baseline it is compared against; and
- important limitations of relying on the pretrained model.

Simply applying a pretrained model to a dataset without substantive ML experimentation or analysis will generally not provide sufficient project scope.

Q4. Can we use LLMs or generative-AI models?

Yes, provided their use supports a substantive machine-learning project.

Open-source models and locally executable tools such as models available through open repositories or frameworks such as **Ollama** may be used. However, a project should not consist primarily of prompting an LLM and reporting its outputs.

If LLMs are used, students should still formulate a clear ML question and conduct meaningful comparisons, evaluation, and error analysis.

Q5. Can we use online or paid APIs?

As a general rule, projects should **not depend on paid or proprietary online APIs** for their core experiments. This includes services where reproducibility depends on continued paid access, private model versions, changing external systems, or substantial API credits.

Open-source and locally runnable alternatives are preferred.

If there is a strong reason to use an external API, the team should discuss it with the teaching team **before making it a central part of the project**. Approval should not be assumed.

Q6. Can we use Kaggle competition datasets or notebooks?

Kaggle datasets may be used and are often suitable for course projects. However, students must conduct their **own analysis, modelling, experiments, and interpretation**.

Public Kaggle notebooks, solutions, or competition pipelines may be consulted only in accordance with the course and Institute academic-integrity policies and must be properly acknowledged where applicable.

A project that substantially reproduces an existing public notebook or competition solution without meaningful additional work will not be considered sufficient.

Q7. Do we need to collect our own dataset?

No. In fact, using an existing, well-documented dataset is generally preferred.

New dataset collection can take substantial time and often leaves insufficient time for modelling and analysis. Collect data only when it is genuinely necessary for the project and after discussing feasibility with the teaching team.

Q8. Can we work on a dataset that has already been heavily preprocessed?

Yes. However, you are still expected to understand the dataset and perform appropriate exploratory analysis.

You should be able to explain what the features and targets represent, how the data was obtained or processed, important distributions or imbalances, possible sources of leakage, and limitations of the dataset.

Q9. How many models must we implement?

There is no fixed required number.

The goal is to perform **meaningful comparisons**, not to create a catalogue of algorithms. Most projects should have at least one sensible baseline and one or more justified alternatives.

Three carefully chosen models with thoughtful analysis are usually more valuable than ten models compared superficially.

Q10. Do we need to implement algorithms from scratch?

Not necessarily. Standard implementations from libraries such as `scikit-learn` or appropriate deep-learning frameworks may be used.

However, students must understand what the methods do, the important hyperparameters and assumptions, and how the reported results were obtained. Library calls do not replace methodological understanding.

Where implementing a component from scratch meaningfully supports the learning objective or project contribution, students are encouraged to do so.

Q11. Is achieving the highest accuracy or best benchmark score important?

No. Project assessment is not a leaderboard competition.

Correct methodology, thoughtful experimental design, appropriate baselines, meaningful analysis, and justified conclusions are more important than obtaining the highest numerical score.

A well-analysed negative result can be a strong project outcome.

Q12. What if our proposed method does not improve over the baseline?

That is acceptable.

You should investigate and explain the result. For example:

- Was the baseline already strong?
- Was the dataset too small or noisy?
- Did the proposed method make inappropriate assumptions?
- Were some features or components ineffective?
- Did the method help only particular cases or subgroups?

A careful analysis of why an approach failed can demonstrate more understanding than an unexplained improvement.

Q13. Can we reproduce an existing research paper?

Yes, but reproduction alone is generally not sufficient.

You should add meaningful work, for example by:

- applying the method to another application or dataset;
- adding additional baselines;
- conducting ablation or sensitivity studies;
- analysing failure modes;
- testing robustness or generalization; or
- critically examining an assumption or reported claim.

Q14. Can different groups work on the same dataset or similar problem?

Yes, unless the teaching team announces otherwise.

Using the same dataset does not imply that the projects are identical. Groups should develop their own problem formulation, methodology, experiments, and analysis.

Substantial sharing of code, reports, experimental results, or project solutions across groups is not permitted.

Q15. Does every group member receive the same marks?

Not necessarily.

The project is a group activity, but individual marks may differ based on demonstrated contribution, intra-group evaluation, understanding during presentations or discussions, and the consistency between claimed and demonstrated work.

Every student is expected to understand the overall project, not only the component they personally implemented.

Q16. Can we use generative AI or coding assistants to help write code or the report?

Only as permitted under the course AI-use policy.

Students remain fully responsible for any generated code, explanations, references, experiments, and claims. Any substantive use should be disclosed where required.

During presentations or project discussions, students may be asked to explain, modify, debug, or justify any part of the submitted work. Material that a student cannot explain should not be submitted as their own work.

Q17. Can our project become a research publication?

Possibly, but publication is not required for the course.

A strong course project may provide a useful starting point for further research. Publication usually requires additional novelty, experiments, analysis, writing, and often domain collaboration beyond what is expected for full marks in AI201.

Students interested in continuing a project toward publication should discuss this with the instructors early.